

Compact VLMs with competitive performance, dramatically lower memory footprints, and fully open-source deployment artifacts.

SmolVLM-256M RAM, batch=1

benchmarks evaluated

2.2B average score

Large VLMs Are Powerful but Impractical for Edge Deployment

Memory footprints and model scale constrain mobile and low-power inference.

EXISTING EFFICIENT OS BASELINE

Molmo-A1B-7B

27.7 GB

GPU RAM requirement

Useful for comparison, but far beyond typical smartphone / edge memory budgets.

LEGACY LARGE VLM

Idefics-80B

300×

larger than SmolVLM-256M

SmolVLM-256M surpasses it despite an 18-month development gap.

TARGET ENVELOPE

On-device VLMs

<1 GB

batch=1 inference target

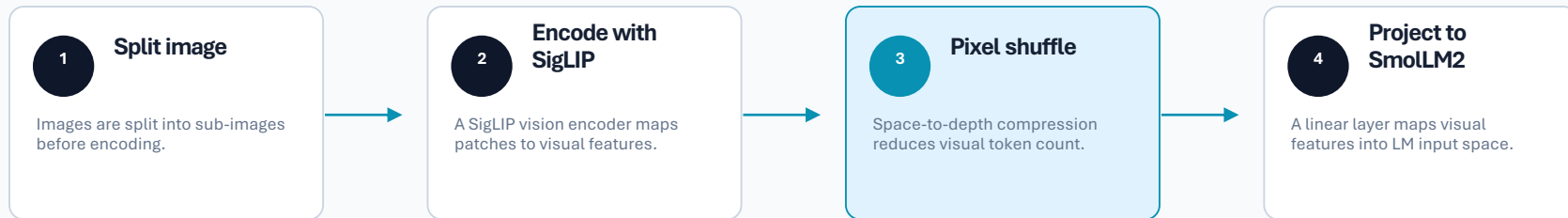
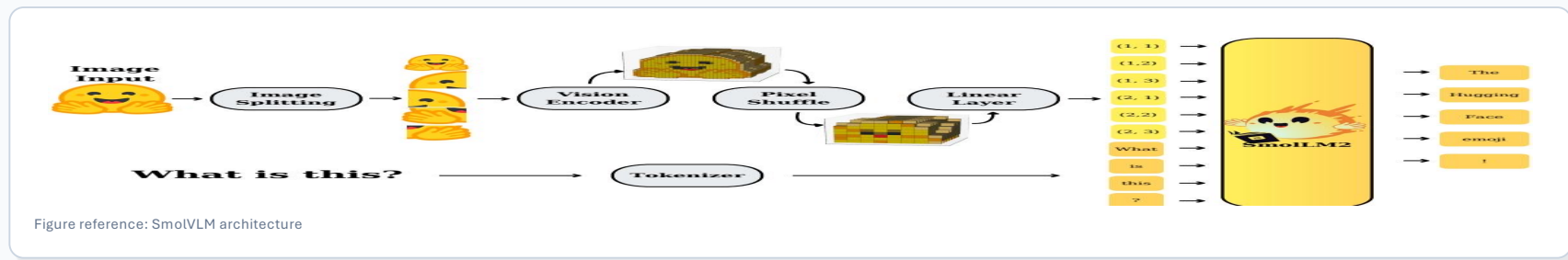
SmolVLM reframes the design goal around architecture efficiency rather than raw scale.

Problem

State-of-the-art VLM capability is not enough: deployment requires memory-aware architecture, compressed visual tokens, and scaled context.

SmolVLM Architecture: Image Splitting, Pixel Shuffle, and SmolLM2 Backbone

A compact multimodal pipeline concatenates compressed visual embeddings with text embeddings.



VIDEO PATH

Frames are sampled and processed through the same image pipeline.

Design levers: encoder-LM balance • token compression • context scaling

Smaller Vision Encoders Outperform Larger Ones in Compact VLMs

Finding 1: balanced encoder-LM parameter allocation matters more than using the largest encoder.

135M LM PAIRING

Experimental comparison from the paper
relative suitability for smallest LMs

93M SigLIP-B/16

OUTPERFORMS

428M SigLIP-SO400M

larger encoder

No numeric accuracy values are provided in the brief; the visual encodes the reported ordering.

INTERPRETATION

93M

428M

SigLIP-B/16 vision encoder SigLIP-SO400M encoder

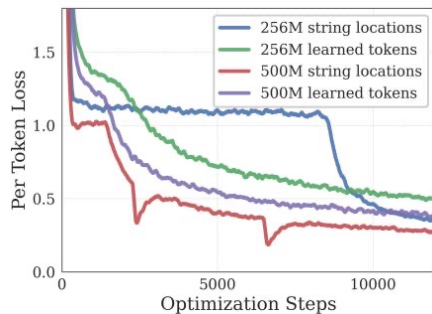
Compact VLMs should not spend most parameters on the vision side. The larger encoder only justifies its ~10% parameter overhead at 1.7B LM scale.

Principle: match encoder capacity to LM scale

Aggressive Pixel Shuffle (r=4) Cuts Visual Tokens 16× for Small Models

Finding 3: space-to-depth compression lowers attention overhead and improves long-context modeling.

MECHANISM



Pixel shuffle rearranges spatial features into channels.

TOKEN MATH

visual tokens

$$\div r^2$$



$$r=4$$

16× fewer visual tokens

SCALE-DEPENDENT RESULT

Small VLMs

prefer $r=4$

Larger models

prefer $r=2$

Aggressive compression is especially valuable when LM capacity and context budget are tight.

Attention overhead ↓

Extending Context from 2K to 16K Tokens Unlocks Higher Image Resolution

Finding 2: longer context improves performance by making room for more visual evidence.



ROPE SCALING

10K → 273K

RoPE base adjustment

FINE-TUNING DATA

**mixed long / short
context**

preserves short-context behavior
while extending resolution capacity

QUALITATIVE IMPACT

**higher image
resolution**

more visual tokens fit without
exceeding context budget

Context is a deployment constraint, not just a text-model setting

SmolVLM Achieves Competitive Performance at Sub-1GB to 4.9GB RAM

Full comparison across single-image, multi-task, and video benchmarks from the source brief

Capability	Benchmark	256M	500M	2.2B	Best Efficient OS
Single-Image	OCRBench	52.6%	61.0%	72.9%	54.7% (Molmo-A1B-7B)
Single-Image	AI2D	46.4%	59.2%	70.0%	71.0% (Molmo-A1B-7B)
Single-Image	ChartQA	55.6%	62.8%	68.7%	48.0% (Molmo-A1B-7B)
Single-Image	TextVQA	50.2%	60.2%	73.0%	61.5% (Molmo-A1B-7B)
Single-Image	DocVQA	58.3%	70.5%	80.0%	77.7% (Molmo-A1B-7B)
Single-Image	ScienceQA	73.8%	80.0%	89.6%	87.5% (Molmo-A1B-7B)
Multi-task	MMMUI	29.0%	33.7%	42.0%	33.9% (Molmo-A1B-7B)
Multi-task	MathVista	35.9%	40.1%	51.5%	37.6% (Molmo-A1B-7B)
Multi-task	MMStar	34.6%	38.3%	46.0%	43.1% (Molmo-A1B-7B)
Video	Video-MME	33.7%	42.2%	52.1%	45.0% (InternVL2-2B)
Video	MLVU	40.6%	47.3%	55.2%	48.2% (InternVL2-2B)
Average	All benchmarks	44.0%	51.0%	59.8%	—
RAM	batch=1	0.8GB	1.2GB	4.9GB	27.7GB (Molmo-A1B-7B)

SOURCE: BRIEF BENCHMARK TABLE

0.8GB

27.7GB

SmolVLM-256M RAM

Molmo-A1B-7B RAM

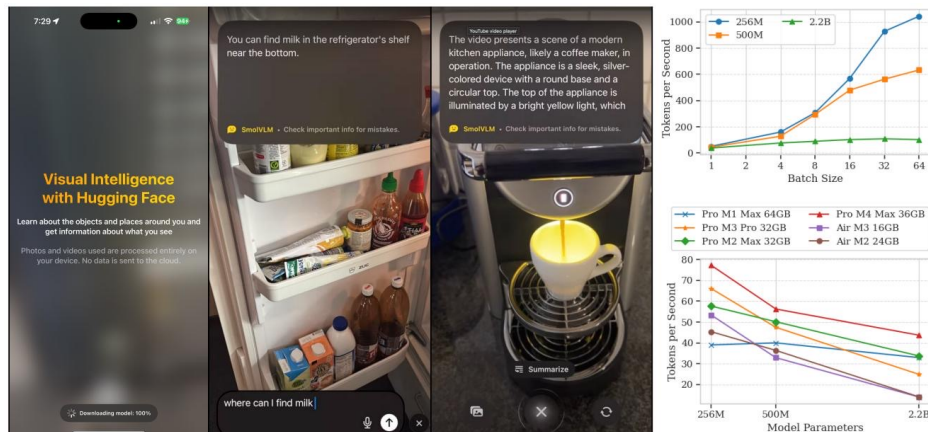
SmolVLM-256M also surpasses the 300× larger Idefics-80B per the paper key findings.

READING THE TABLE

- SmolVLM-2.2B reaches 59.8% average.
- 256M and 500M operate below 1.2GB RAM.
- Video baselines come from InternVL2-2B; other efficient OS rows use Molmo-A1B-7B.

Real-Time Inference on iPhones: Up to 80 Tokens/Second on Apple Silicon

HuggingSnap runs photos and videos entirely locally with no cloud dependency.



Local photos + videos • no cloud dependency

Deployment implication: SmolVLM shifts VLM inference from server-class GPUs toward mobile and low-power edge hardware.

Key Takeaways: Design Principles for Efficient Multimodal Models

SmolVLM shows that careful architecture can beat brute-force scale for edge VLMs.

01

Balance encoder-LM
compute

Small LMs benefit from compact encoders;
larger encoders only pay off near 1.7B LM scale.

02

Compress visual tokens
aggressively

Use $r=4$ pixel shuffle for small VLMs to cut
visual tokens 16x.

03

Extend context
deliberately

Scale from 2K toward 8K/16K with RoPE base
10K → 273K and mixed-context fine-tuning.

04

Optimize for RAM
envelope

Sub-1GB inference is achievable: SmolVLM-
256M uses 0.8GB at batch=1.

05

Open the deployment
stack

Weights, datasets, code, and the HuggingSnap
mobile app are fully open-sourced.

Implication: efficient multimodal models need architecture co-design across vision, tokens, context, and deployment memory.